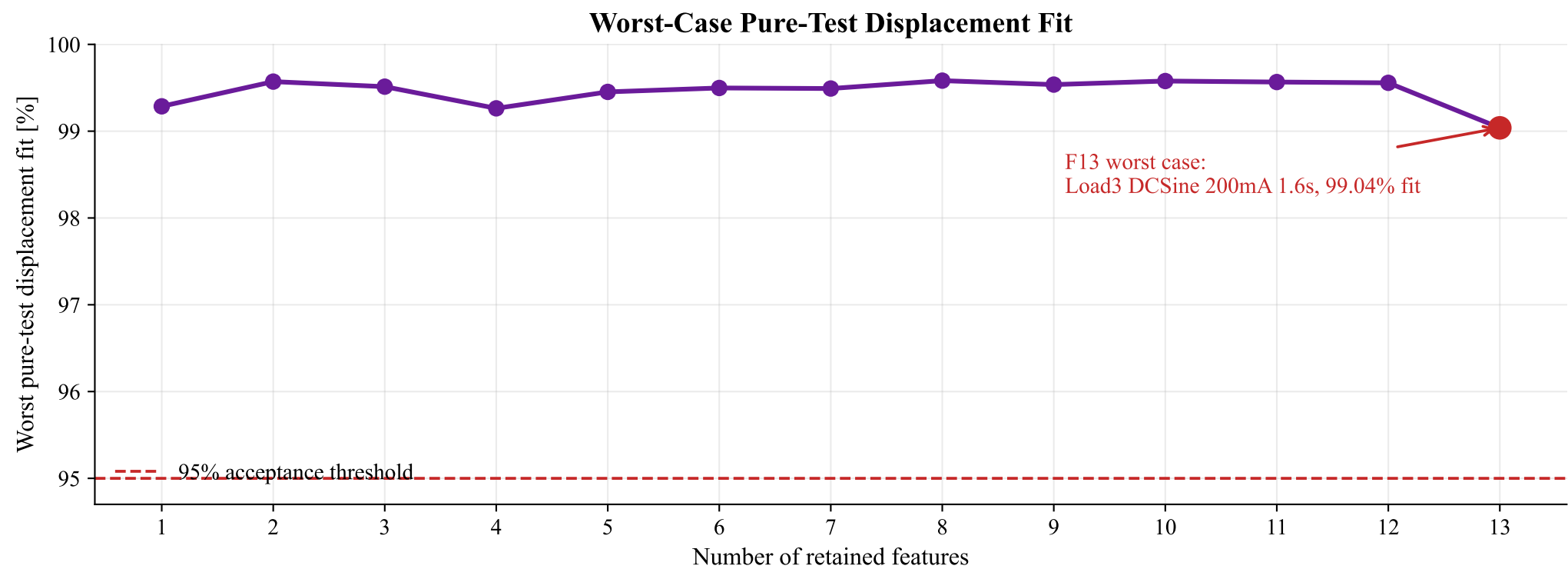
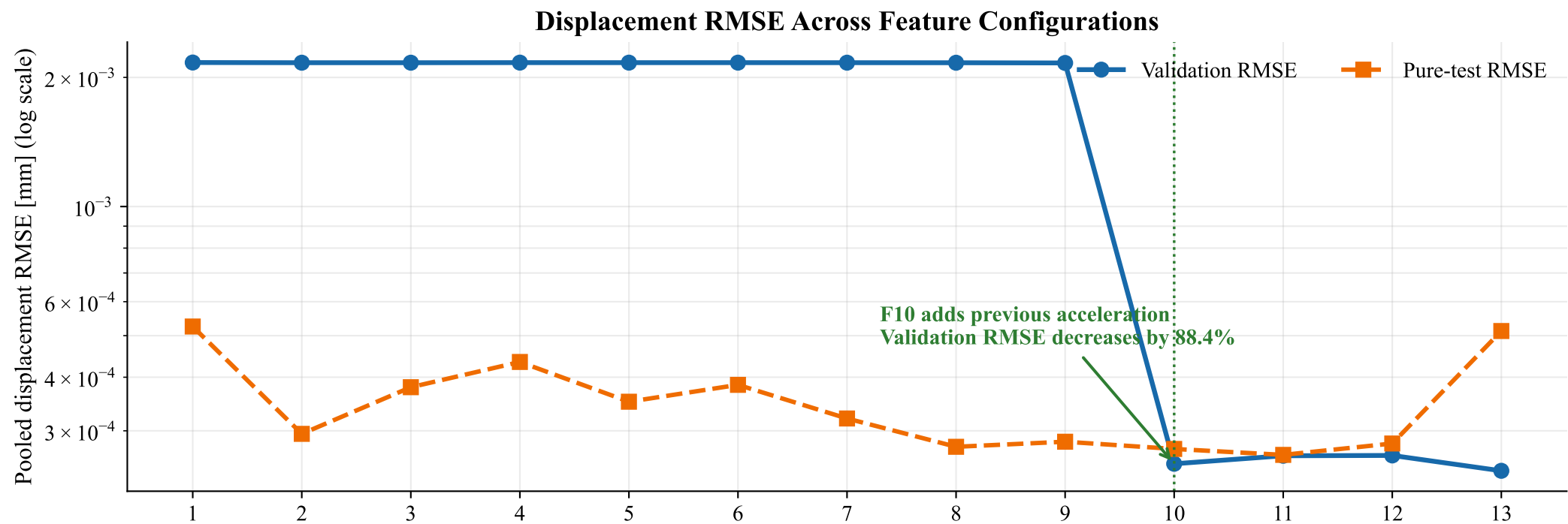


Cumulative LSTM Feature-Ablation Results



Summary

Largest improvement

F09 → F10
+ previous acceleration
88.4% lower validation RMSE

Best pure-test RMSE

F11: 0.000264 mm
Descriptive confirmation

Best validation RMSE

F13: 0.000242 mm
Validation-selected model

Generalization observation

F13 pure-test RMSE is
94.4% higher than F11

All cases use the same split and one random seed. Feature selection must use validation data; pure tests are final confirmation. Force is excluded because it is computed by a separate causal rule.